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SOTWT $\rightarrow$ Stationary discrete-time
wavelet transform

2D version $\rightarrow$ can also be used
for time frequency
domain, low and
high frequencies
are related to
smooth variation
and borders

$\hookrightarrow$ for filtering, we could apply
bandpass thresholds
in the whole part of a
motion

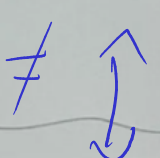
$\hookrightarrow$ long exposure is pretty
much the same, you get
more or less. As such as you
modify the pattern will appear
related to the function. Then
pattern any setting like digit
change in the color or the phase
shift

$\hookrightarrow$ it can also be used for
feature extraction

if we apply our transformation in SE, we shift the result and apply it again we expect the result to be shifted as well (this is a linear system).

$x[n] = \{2, 3, -4, 5, 1, 5, 3, -2\}$
 $DTWT_3(x[n]) \stackrel{\text{Haar}}{=} \left\{ \underbrace{\frac{5}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{6}{\sqrt{2}}, \frac{1}{\sqrt{2}}}_{\text{Trend}} \middle| \underbrace{\frac{-1}{\sqrt{2}}, \frac{-9}{\sqrt{2}}, \frac{-4}{\sqrt{2}}, \frac{5}{\sqrt{2}}}_{\text{fluctuation}} \right\}$

$x_s[n] = \{-2, 2, 3, -4, 5, 1, 5, 3\} \rightarrow x[n] \text{ right-shifted by one position}$
 $DTWT_3(x_s[n]) \stackrel{\text{Haar}}{=} \left\{ \underbrace{\frac{0}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, \frac{6}{\sqrt{2}}, \frac{8}{\sqrt{2}}}_{\text{Trend}} \middle| \underbrace{\frac{-4}{\sqrt{2}}, \frac{7}{\sqrt{2}}, \frac{4}{\sqrt{2}}, \frac{2}{\sqrt{2}}}_{\text{fluctuation}} \right\}$



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this is not a linear system invariant in time, because they are different

$\therefore$ DTWT is shift variant

SDTWT $\rightarrow$ we remove the downsampling
 by doing that the signal will get
 larger $\rightarrow$ a way to do that is
 using the Mallat's algorithm
 but instead of shifting sum
 of nodes by 2, we shift by 1
 in one of the trees is for
 downsampling. Now we need
 need to adopt a bunch of
 things

$\rightarrow$ To find that we instead
 do

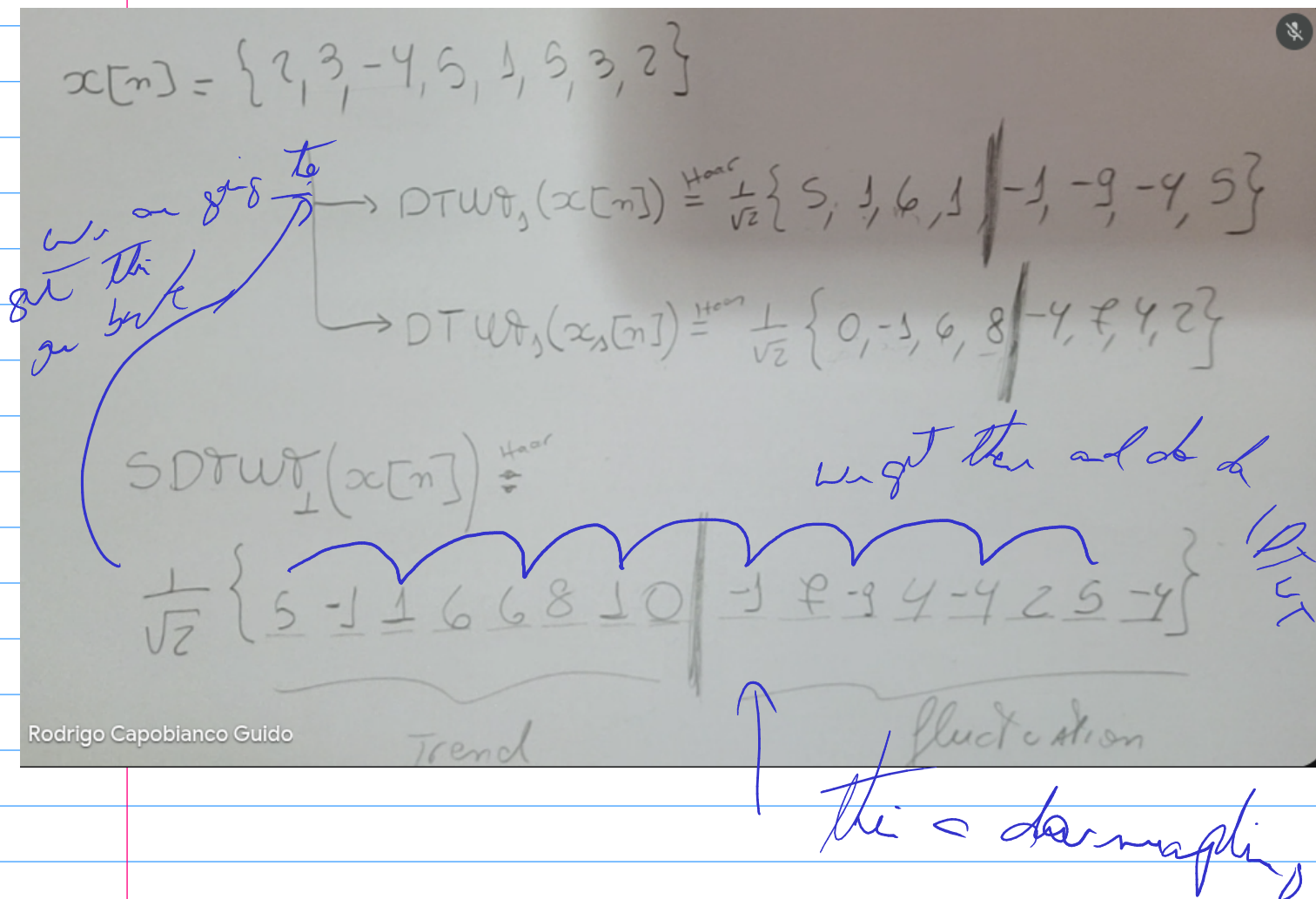
$x[n] = \{2, 3, -4, 5, 1, 5, 3, 2\}$
 regular $\rightarrow$ $\text{DTWT}_3(x[n]) \stackrel{\text{Haar}}{=} \frac{1}{\sqrt{2}} \{5, 1, 6, 1 \mid -1, -9, -4, 5\}$
 lifted $\rightarrow$ $\text{DTWT}_3(x_s[n]) \stackrel{\text{Haar}}{=} \frac{1}{\sqrt{2}} \{0, -1, 6, 8 \mid -4, 4, 4, 2\}$
 $\text{SDTWT}_1(x[n]) \stackrel{\text{Haar}}{=} \frac{1}{\sqrt{2}} \{5, -1, 1, 6, 6, 8, 1, 0 \mid -1, 4, -9, 4, -4, 2, 5, -4\}$
 Trend fluctuation

$\uparrow$
 it double of size

for the model level we want that
rigid and do the same process

the same thing can be done in
its vector version, which depends in size

to get the inner



SOT WT can be defined by CD,
but it's not enough, since it would
increase the image size.